RH_CHECKPOINT – Full Technical Master v3.3

Transport–Spectral Contradiction Program for the Riemann Hypothesis

Closure, Drift, No-Hiding, Packet Coercivity, Gaussian Regularization, and Model Bricks

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Abstract

We present a sharpened localized transport–spectral checkpoint for the Riemann Hypothesis. The framework isolates a precise conditional contradiction inside a dipole-regularized residue channel localized around a single candidate off-critical zero

$$\rho = \beta + i\gamma, \qquad \beta \neq \frac{1}{2}.$$

On the closure side, a localized exact window law forces the localized excess energy to vanish:

$$E_{\mathrm{ex}}^{(\rho)}(f) = 0.$$

On the obstruction side, a transported drift law

$$\delta_\rho(u) = a_\rho u + O(1), \qquad a_\rho = \frac{\beta - \frac{1}{2}}{2\pi} \neq 0,$$

generates a quartic penalty

$$P_4(\rho; \tau, L) \gtrsim \tau^4 L.$$

If the localized witness bank detects this drift coercively and cross/remainder leakage remains subcritical, then the same localized channel satisfies

$$E_{\mathrm{ex}}^{(\rho)}(f) > 0,$$

contradicting closure.

This v3.2 master version keeps the main proof chain short but preserves all essential technical details: Fourier preliminaries, dipole regularization, localized witness banks, exact-window closure, quartic obstruction, bridge inequality, bank-subcritical remainder, finite Gaussian Gram positivity, windowed packet coercivity, perturbative stability, nonstationary cross-term decay, Gaussian heat-semigroup regularization, and the Source–Love Channel as a separate meta-model.

The document remains strictly conditional. It is not an unconditional proof of RH. It reduces RH to explicit analytic bottlenecks on the genuine zeta residue channel and supplies rigorous model bricks showing that the required mechanisms are mathematically nonempty.

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1 Public Clarification

Important clarification. This document is a conditional checkpoint, not an unconditional proof of the Riemann Hypothesis. It isolates a precise contradiction mechanism and reduces RH to explicit analytic tasks that must still be verified on the genuine zeta residue channel. All model results in the appendices are proved only in controlled regimes; their transfer to the actual explicit-formula channel remains open.

Důležité upřesnění. Tento dokument není nepodmíněný důkaz Riemannovy hypotézy. Jde o podmíněný checkpoint, který izoluje přesný mechanismus rozporu a převádí RH na explicitní analytické úkoly na skutečném zeta-reziduálním kanálu. Všechny modelové výsledky v dodatcích jsou dokázány pouze v kontrolovaných modelech; jejich přenos na skutečný kanál explicitní formule zůstává otevřený.

Remark 1.1. *This clarification is part of the logical status of the paper. The model appendices are not promoted to genuine zeta-side theorems. The RH implication requires verifying the corresponding estimates for the actual residue channel.*

2 Introduction

The Riemann Hypothesis asserts that every nontrivial zero

$$\rho = \beta + i\gamma$$

of the Riemann zeta function satisfies

$$\beta = \frac{1}{2}.$$

The present framework does not attempt to locate all zeros simultaneously. Instead, it asks a localized question:

Would the existence of even a single off-critical zero necessarily create a detectable energetic obstruction in a suitably localized readout channel?

The intended contradiction has two sides:

$$\boxed{\text{localized exact window law} \Longrightarrow E_{\mathrm{ex}}^{(\rho)}(f) = 0}$$

and

$$\boxed{\text{transported drift + witness coercivity + no-hiding} \Longrightarrow E_{\mathrm{ex}}^{(\rho)}(f) > 0.}$$

Thus the program reduces RH to the verification of a localized exact window law and a localized bridge package for every candidate off-critical zero-channel.

The document is organized as follows. The main body gives definitions, assumptions, and the fixed-channel contradiction. The appendices contain the model bricks: soft readout survival, pure exponential coercivity, windowed packet coercivity, cross-term decay, Gaussian regularization, and the Source–Love Channel meta-model.

3 Why Localization Matters

Localization provides three advantages over a purely global approach.

(i) **Single-channel obstruction.** A candidate zero ρ is treated as an individual transported mode, rather than as an anonymous contribution to the global zero set.

(ii) **Explicit interference control.** The interaction between the candidate mode and the remainder becomes a localized cross term that can be attacked by nonstationary-phase estimates.

(iii) **Modular bottlenecks.** The proof decomposes into independent analytic tasks: transported drift, witness coercivity, no-hiding leakage, bank-subcritical remainder, Gram stability, exact window closure, and model-to-zeta transfer.

Localization does not weaken the statement. It sharpens the target.

4 Public-Facing Synthesis, Made Proof-Safe

4.1 Compressed description

If an off-critical zero exists, then after suitable transport it should generate a localized linear drift. A witness bank should detect this drift as a coercive quartic obstruction. If remainder and cross terms are subcritical, the localized excess becomes positive. A localized exact window law would force the same excess to vanish. The contradiction excludes the off-critical channel once all assumptions are verified on the genuine zeta residue side.

In one line:

$$\text{localize the channel} \longrightarrow \text{read the drift} \longrightarrow \text{keep leakage subcritical} \longrightarrow \text{close the window.}$$

4.2 Claim hygiene

H1. Do not present the framework as an unconditional proof of RH.

H2. Do not promote model lemmas to statements about the genuine zeta residue channel.

H3. Do not treat visibility, screenshots, or social reception as mathematical evidence.

H4. Keep the proof chain fixed:

$$\text{drift} + \text{coercivity} + \text{no-hiding leakage} + \text{exact window} \Longrightarrow \text{contradiction.}$$

H5. Separate exposition and proof components. Symbolic language may guide intuition but cannot replace estimates.

5 Dependency Ledger

The document separates four logical layers.

Layer	Content	Status
Definitions	localized bank, excess, window, drift, penalty	structural
Algebraic implications	closure, bridge inequality, fixed-channel contradiction	proved here
Model bricks	soft survival, packet coercivity, cross-term decay, Gaussian hammer	proved in model
Zeta-side tasks	drift law, coercivity, leakage, Gram bounds, exact window	open

The proof chain is not

$$\text{model lemma} \Longrightarrow \text{RH.}$$

It is

$$\text{model lemma} \Longrightarrow \text{evidence that the analytic target is nonempty.}$$

The actual RH implication requires corresponding statements on the genuine zeta residue channel.

6 Assumption Ledger for the Fixed Channel

Fix a candidate zero

$$\rho = \beta + i\gamma, \qquad \beta \neq \frac{1}{2}.$$

The fixed-channel contradiction uses the following assumptions.

A1. Transported drift. There exists a transported phase/readout variable

$$\delta_\rho(u) = a_\rho u + r_\rho(u), \qquad a_\rho = \frac{\beta - \frac{1}{2}}{2\pi} \neq 0,$$

where r_ρ is bounded.

A2. Localized exact window law. The localized witness-bank energy satisfies

$$Q^{(\rho)}_{\text{bank}}(f;\tau,L) = E^{(\rho)}_0 L + J^{(\rho)}_f(\tau+L) - J^{(\rho)}_f(\tau) + R^{(\rho)}(\tau,L),$$

with $J^{(\rho)}_f$ bounded and

$$\lim_{L\to\infty} \sup_{\tau\in\mathbb{R}} \frac{|R^{(\rho)}(\tau,L)|}{L} = 0.$$

A3. Localized witness coercivity. There exist $\kappa > 0$ and $C_0 \geq 0$ such that

$$Q^{(\rho)}_\rho(\tau,L) \geq \kappa P_4(\rho;\tau,L) - C_0 L.$$

A4. Localized no-hiding leakage. There exist $C_1 \geq 0$ and $\eta(\tau) \to 0$ such that

$$Q^{(\rho)}_{\text{cross}}(\tau,L) + Q^{(\rho)}_{\text{rest}}(\tau,L) \geq -\eta(\tau)P_4(\rho;\tau,L) - C_1 L.$$

Only A1–A4 enter the fixed-channel contradiction. The model-to-zeta transfer remains open.

7 Fourier Preliminaries

We use the Fourier transform convention

$$\widehat{f}(\xi) = \int_{\mathbb{R}} f(x) e^{-2\pi i x\xi}\, \mathrm{d}x.$$

Let

$$\psi(x) = e^{-\pi x^2}.$$

Lemma 7.1 (Gaussian self-duality)**.** *With the above convention,*

$$\widehat{\psi} = \psi.$$

Proof. Let

$$F(\xi) = \widehat{\psi}(\xi).$$

Differentiating under the integral sign,

$$F'(\xi) = \int_{\mathbb{R}} e^{-\pi x^2}(-2\pi i x) e^{-2\pi i x\xi}\, \mathrm{d}x.$$

Since

$$\frac{\mathrm{d}}{\mathrm{d}x} e^{-\pi x^2} = -2\pi x e^{-\pi x^2},$$

we get

$$F'(\xi) = i \int_{\mathbb{R}} \frac{\mathrm{d}}{\mathrm{d}x}(e^{-\pi x^2}) e^{-2\pi i x\xi}\, \mathrm{d}x.$$

Integration by parts gives

$$F'(\xi) = -2\pi\xi F(\xi).$$

Thus

$$F(\xi) = Ce^{-\pi\xi^2}.$$

At $\xi = 0$,

$$C = F(0) = \int_{\mathbb{R}} e^{-\pi x^2}\, \mathrm{d}x = 1.$$

Therefore

$$F(\xi) = e^{-\pi\xi^2}.$$

□

Since the Fourier transform satisfies

$$\mathcal{F}^4 = I$$

on $L^2(\mathbb{R})$, its spectral values lie in

$$\{1, -i, -1, i\}.$$

For each

$$\lambda \in \{1, -i, -1, i\},$$

define

$$P_\lambda = \frac{1}{4}\sum_{m=0}^{3} \lambda^{-m}\mathcal{F}^m.$$

Proposition 7.2 (Fourier spectral projectors)**.** *For every $\lambda \in \{1, -i, -1, i\}$,*

$$P_\lambda^2 = P_\lambda, \qquad \mathcal{F}(P_\lambda f) = \lambda P_\lambda f.$$

In particular,

$$P_{+1} = \frac{1}{4}(I + \mathcal{F} + \mathcal{F}^2 + \mathcal{F}^3)$$

projects onto the Fourier-fixed self-dual sector.

Proof. The identity $\mathcal{F}^4 = I$ implies that the minimal polynomial of $\mathcal{F}$ divides

$$z^4 - 1.$$

The operators P_λ are the finite spectral projectors for the cyclic group generated by $\mathcal{F}$.

Directly,

$$\mathcal{F}P_\lambda = \frac{1}{4}\sum_{m=0}^{3} \lambda^{-m}\mathcal{F}^{m+1}.$$

Reindexing modulo 4, and using $\lambda^4 = 1$, gives

$$\mathcal{F}P_\lambda = \lambda P_\lambda.$$

For idempotence, use finite Fourier orthogonality:

$$\frac{1}{4}\sum_{m=0}^{3} (\lambda^{-1}\mu)^m = \begin{cases} 1, & \lambda = \mu, \\ 0, & \lambda \neq \mu. \end{cases}$$

Hence $P_\lambda P_\mu = 0$ for $\lambda \neq \mu$, and $P_\lambda^2 = P_\lambda$. □

Remark 7.3 (Fourier as phase accounting)**.** *In the Hermite basis $\{h_n\}_{n\geq 0}$,*

$$\mathcal{F}h_n = (-i)^n h_n.$$

Thus Fourier evolution is pure phase cycling on Hermite modes:

$$1,\ -i,\ -1,\ i.$$

In the correct basis, Fourier does not mix structure chaotically. It rotates phase mode by mode.

8 Dipole-Regularized Residue Channel

Let μ denote the residue distribution arising from a fixed explicit-formula normalization. Let

$$\psi \in \mathcal{S}(\mathbb{R})$$

be a Schwartz test function satisfying the dipole condition

$$\widehat{\psi}(0) = 0.$$

Define the dipole-regularized residue channel

$$f = \psi * \mu.$$

The dipole condition removes the constant component and leaves a fluctuation channel suitable for second-difference witness tests.

For $a > 0$, define the second-difference witness operator

$$(W_a h)(u) = h(u+a) - 2h(u) + h(u-a).$$

Remark 8.1. *The operator W_a annihilates affine functions and detects curvature at scale a. This is why it is suited to measuring drift-induced obstruction.*

9 Localized Witness Bank

Fix a candidate zero

$$\rho = \beta + i\gamma.$$

Let $w_\rho(u) \geq 0$ be a localization window. Let

$$a_j > 0, \qquad j = 0, \ldots, J,$$

be finitely many witness scales, and let $w_j > 0$ be positive bank weights.

Definition 9.1 (Localized witness bank)**.** *The ρ-localized witness-bank energy is*

$$Q^{(\rho)}_{\text{bank}}(f;\tau,L) = \sum_{j=0}^{J} w_j^2 \int_\tau^{\tau+L} w_\rho(u)|W_{a_j} f(u)|^2 \, \mathrm{d}u.$$

Definition 9.2 (Localized excess)**.** *Given a baseline density $E_0^{(\rho)} \geq 0$, define*

$$E^{(\rho)}_{\text{ex}}(f) = \limsup_{L\to\infty} \sup_{\tau\in\mathbb{R}} \frac{Q^{(\rho)}_{\text{bank}}(f;\tau,L) - E_0^{(\rho)} L}{L}.$$

Assume a single-zero splitting

$$f = f_\rho + f_{\text{rest}}.$$

Then

$$W_{a_j} f = W_{a_j} f_\rho + W_{a_j} f_{\text{rest}}.$$

Therefore

$$Q^{(\rho)}_{\text{bank}}(f;\tau,L) = Q^{(\rho)}_{\rho}(\tau,L) + Q^{(\rho)}_{\text{cross}}(\tau,L) + Q^{(\rho)}_{\text{rest}}(\tau,L),$$

where

$$Q^{(\rho)}_{\rho}(\tau,L) = \sum_{j=0}^{J} w_j^2 \int_\tau^{\tau+L} w_\rho(u)|W_{a_j} f_\rho(u)|^2 \, \mathrm{d}u,$$

$$Q^{(\rho)}_{\text{cross}}(\tau,L) = 2\sum_{j=0}^{J} w_j^2 \operatorname{Re} \int_\tau^{\tau+L} w_\rho(u) W_{a_j} f_\rho(u) \overline{W_{a_j} f_{\text{rest}}(u)} \, \mathrm{d}u,$$

and

$$Q^{(\rho)}_{\text{rest}}(\tau,L) = \sum_{j=0}^{J} w_j^2 \int_\tau^{\tau+L} w_\rho(u)|W_{a_j} f_{\text{rest}}(u)|^2 \, \mathrm{d}u.$$

10 Localized Exact Window Law

Definition 10.1 (Localized exact window law)**.** *We say that the ρ-localized exact window law holds if there exist a bounded function*

$$J_f^{(\rho)} : \mathbb{R} \to \mathbb{R}$$

and a remainder $R^{(\rho)}(\tau, L)$ such that

$$Q_{\text{bank}}^{(\rho)}(f; \tau, L) = E_0^{(\rho)} L + J_f^{(\rho)}(\tau + L) - J_f^{(\rho)}(\tau) + R^{(\rho)}(\tau, L),$$

where

$$\sup_{\tau \in \mathbb{R}} |J_f^{(\rho)}(\tau)| < \infty,$$

and

$$\lim_{L \to \infty} \sup_{\tau \in \mathbb{R}} \frac{|R^{(\rho)}(\tau, L)|}{L} = 0.$$

Theorem 10.2 (Localized closure)**.** *If the localized exact window law holds, then*

$$E_{\text{ex}}^{(\rho)}(f) = 0.$$

Proof. Divide the exact window identity by L:

$$\frac{Q_{\text{bank}}^{(\rho)}(f; \tau, L) - E_0^{(\rho)} L}{L} = \frac{J_f^{(\rho)}(\tau + L) - J_f^{(\rho)}(\tau)}{L} + \frac{R^{(\rho)}(\tau, L)}{L}.$$

Taking absolute values and then the supremum over τ,

$$\sup_{\tau} \left| \frac{Q_{\text{bank}}^{(\rho)}(f; \tau, L) - E_0^{(\rho)} L}{L} \right| \leq \frac{2\|J_f^{(\rho)}\|_\infty}{L} + \sup_{\tau} \frac{|R^{(\rho)}(\tau, L)|}{L}.$$

Both terms tend to 0 as $L \to \infty$. Hence

$$E_{\text{ex}}^{(\rho)}(f) = 0.$$

□

11 Transported Drift and Quartic Obstruction

Definition 11.1 (Transported drift law)**.** *For a candidate zero*

$$\rho = \beta + i\gamma,$$

we say that the transported drift law holds if

$$\delta_\rho(u) = a_\rho u + r_\rho(u), \qquad a_\rho = \frac{\beta - \frac{1}{2}}{2\pi},$$

where r_ρ is bounded.

Definition 11.2 (Quartic penalty)**.** *Define*

$$P_4(\rho; \tau, L) = \int_\tau^{\tau+L} |\delta_\rho(u)|^4 \, \mathrm{d}u.$$

Lemma 11.3 (Quartic lower bound)**.** *Assume the transported drift law and $a_\rho \neq 0$. Then for sufficiently large τ and all $L \geq 1$,*

$$P_4(\rho; \tau, L) \geq c_\rho \tau^4 L$$

for some constant $c_\rho > 0$.

Proof. Write

$$\delta_\rho(u) = a_\rho u + r_\rho(u), \qquad |r_\rho(u)| \le M.$$

For sufficiently large u,

$$|\delta_\rho(u)| \ge |a_\rho| u - M \ge \frac{|a_\rho|}{2} u.$$

Thus for $u \in [\tau, \tau + L]$ and τ sufficiently large,

$$|\delta_\rho(u)|^4 \ge \frac{|a_\rho|^4}{16} u^4 \ge \frac{|a_\rho|^4}{16} \tau^4.$$

Integrating over an interval of length L,

$$P_4(\rho; \tau, L) \ge \frac{|a_\rho|^4}{16} \tau^4 L.$$

□

Lemma 11.4 (Polynomial upper bound for quartic penalty)**.** *Assume the transported drift law and $|r_\rho(u)| \le M$. Then there exists $C_\rho > 0$ such that*

$$P_4(\rho; \tau, L) \le C_\rho \Big((\tau + L)^5 - \tau^5 + L \Big)$$

for all $\tau \ge 1$, $L \ge 1$.

Proof. Since

$$|\delta_\rho(u)| \le |a_\rho| u + M,$$

there is a constant $C > 0$ such that

$$|\delta_\rho(u)|^4 \le C(u^4 + 1).$$

Therefore

$$P_4(\rho; \tau, L) \le C \int_\tau^{\tau+L} (u^4 + 1) \, \mathrm{d}u = C \left(\frac{(\tau + L)^5 - \tau^5}{5} + L \right).$$

Absorb the factor $1/5$ into the constant. □

Remark 11.5 (Why $m_0 < 2$ is critical)**.** *The quartic obstruction grows like*

$$P_4(\rho; \tau, L) \gtrsim \tau^4 L.$$

If a remainder contribution grows like

$$Q_{\mathrm{rest}}^{(\rho)}(\tau, L) \lesssim \tau^{2m_0} L,$$

then

$$\frac{Q_{\mathrm{rest}}^{(\rho)}}{P_4} \lesssim \tau^{2m_0 - 4}.$$

Thus $m_0 < 2$ is exactly the condition $2m_0 < 4$, which forces the ratio to vanish.

12 Localized Bridge Package

Definition 12.1 (Localized witness coercivity)**.** *The localized bank is coercive on f_ρ if there exist $\kappa > 0$ and $C_0 \geq 0$ such that*

$$Q_\rho^{(\rho)}(\tau, L) \geq \kappa P_4(\rho; \tau, L) - C_0 L$$

for sufficiently large τ and all $L \geq 1$.

Definition 12.2 (Localized no-hiding leakage)**.** *The localized leakage is subcritical if there exist $C_1 \geq 0$ and $\eta(\tau) \to 0$ such that*

$$Q_{\text{cross}}^{(\rho)}(\tau, L) + Q_{\text{rest}}^{(\rho)}(\tau, L) \geq -\eta(\tau) P_4(\rho; \tau, L) - C_1 L$$

for sufficiently large τ and all $L \geq 1$.

Proposition 12.3 (Localized bridge inequality)**.** *If localized witness coercivity and localized no-hiding leakage hold, then*

$$Q_{\text{bank}}^{(\rho)}(f; \tau, L) \geq (\kappa - \eta(\tau)) P_4(\rho; \tau, L) - (C_0 + C_1) L.$$

In particular, for sufficiently large τ,

$$Q_{\text{bank}}^{(\rho)}(f; \tau, L) \geq \frac{\kappa}{2} P_4(\rho; \tau, L) - (C_0 + C_1) L.$$

Proof. By the decomposition,

$$Q_{\text{bank}}^{(\rho)} = Q_\rho^{(\rho)} + Q_{\text{cross}}^{(\rho)} + Q_{\text{rest}}^{(\rho)}.$$

Using coercivity and no-hiding leakage,

$$Q_{\text{bank}}^{(\rho)} \geq \kappa P_4 - C_0 L - \eta(\tau) P_4 - C_1 L.$$

Hence

$$Q_{\text{bank}}^{(\rho)} \geq (\kappa - \eta(\tau)) P_4 - (C_0 + C_1) L.$$

Since $\eta(\tau) \to 0$, for sufficiently large τ,

$$\eta(\tau) \leq \frac{\kappa}{2}.$$

The second inequality follows. □

13 Bank-Subcritical Remainder Criterion

The following criterion explains why a C^2-type remainder bound with exponent $m_0 < 2$ is sufficient for bank-subcriticality.

Lemma 13.1 (Second-difference bound)**.** *Let*

$$h \in C^2([\tau - a, \tau + L + a]).$$

Then for every $u \in [\tau, \tau + L]$,

$$|W_a h(u)| \leq a^2 \sup_{v \in [u-a, u+a]} |h''(v)|.$$

Proof. By Taylor's theorem with integral remainder,

$$h(u + a) = h(u) + a h'(u) + \int_0^a (a - s) h''(u + s) \, \mathrm{d}s,$$

and

$$h(u - a) = h(u) - a h'(u) + \int_0^a (a - s) h''(u - s) \, \mathrm{d}s.$$

Adding and subtracting $2h(u)$, the linear terms cancel:

$$W_a h(u) = \int_0^a (a-s)\big(h''(u+s) + h''(u-s)\big)\ \mathrm{d}s.$$

Therefore

$$|W_a h(u)| \le 2\int_0^a (a-s)\ \mathrm{d}s \sup_{v\in[u-a,u+a]} |h''(v)| = a^2 \sup_{v\in[u-a,u+a]} |h''(v)|.$$

□

Proposition 13.2 (Integral bank-subcritical remainder criterion)**.** *Assume that for sufficiently large u,*

$$|f''_{\mathrm{rest}}(u)| \le C(1+u)^{m_0}$$

with

$$m_0 < 2.$$

Assume also that the localization window is bounded:

$$0 \le w_\rho(u) \le M_\rho.$$

Then, for fixed finite bank scales,

$$Q^{(\rho)}_{\mathrm{rest}}(\tau, L) \le C' \int_\tau^{\tau+L} (1+u)^{2m_0}\ \mathrm{d}u.$$

If the quartic obstruction satisfies the integral lower bound

$$P_4(\rho;\tau,L) \ge c\int_\tau^{\tau+L} u^4\ \mathrm{d}u - C_0 \int_\tau^{\tau+L} u^3\ \mathrm{d}u,$$

then, as the transported window moves to large scale,

$$\frac{Q^{(\rho)}_{\mathrm{rest}}(\tau,L)}{P_4(\rho;\tau,L)} \to 0.$$

In particular,

$$Q^{(\rho)}_{\mathrm{rest}} = o(P_4).$$

Proof. By the second-difference bound, for each fixed scale a_j,

$$|W_{a_j} f_{\mathrm{rest}}(u)| \le C_j(1+u)^{m_0}.$$

Therefore

$$|W_{a_j} f_{\mathrm{rest}}(u)|^2 \le C'_j(1+u)^{2m_0}.$$

Multiplying by the bounded localization window, summing over finitely many j, and integrating,

$$Q^{(\rho)}_{\mathrm{rest}}(\tau,L) \le C' \int_\tau^{\tau+L} (1+u)^{2m_0}\ \mathrm{d}u.$$

Since $2m_0 < 4$, the numerator has strictly lower polynomial order than

$$\int_\tau^{\tau+L} u^4\ \mathrm{d}u.$$

The correction

$$\int_\tau^{\tau+L} u^3\ \mathrm{d}u$$

is also lower order relative to the u^4-term at large scale. Hence

$$\frac{Q^{(\rho)}_{\mathrm{rest}}(\tau,L)}{P_4(\rho;\tau,L)} \to 0.$$

□

14 Uniform Gaussian Gram Coercivity Revisited: Failure, Separation, and Jet Repair

This section refines the Gaussian Gram bottleneck. The original formulation asked for a height-uniform lower bound on ordinary Gaussian Gram matrices for local unfolded zero clusters. That formulation is too strong without a separation hypothesis. Near-colliding points force the ordinary Gram matrix to become nearly singular.

The correct replacement is a separation-or-jet principle:

ordinary Gram coercivity for separated modes + confluent jet Gram coercivity for near-collisions.

This does not solve the genuine zeta-side Gram problem. It clarifies the correct analytic target.

14.1 No-go lemma without separation

Lemma 14.1 (Failure of uniform ordinary Gram coercivity without separation)**.** *Fix $\alpha > 0$. There is no positive constant $c > 0$ such that*

$$\lambda_{\min}(K) \geq c$$

for all finite configurations of distinct points, where

$$K_{jk} = e^{-\alpha(x_j - x_k)^2}.$$

Already for two points, the smallest eigenvalue can be made arbitrarily small.

Proof. Take two points

$$x_1 = 0, \qquad x_2 = \varepsilon.$$

Then the Gaussian Gram matrix is

$$K_\varepsilon = \begin{pmatrix} 1 & e^{-\alpha\varepsilon^2} \\ e^{-\alpha\varepsilon^2} & 1 \end{pmatrix}.$$

Its eigenvalues are

$$\lambda_+(\varepsilon) = 1 + e^{-\alpha\varepsilon^2}, \qquad \lambda_-(\varepsilon) = 1 - e^{-\alpha\varepsilon^2}.$$

As $\varepsilon \to 0$,

$$\lambda_-(\varepsilon) = 1 - e^{-\alpha\varepsilon^2} \sim \alpha\varepsilon^2 \to 0.$$

Thus

$$\inf_\varepsilon \lambda_{\min}(K_\varepsilon) = 0.$$

Therefore a uniform positive lower bound cannot hold without an additional separation condition or a different readout resolving near-collisions. □

Remark 14.2 (Meaning for the RH checkpoint)**.** *The original ordinary Gram coercivity target must not be stated without qualification. Local clusters with arbitrarily small gaps make the ordinary Gaussian atoms nearly linearly dependent. This is not a failure of the bridge architecture; it identifies the correct refinement needed for the no-hiding mechanism.*

14.2 Separated Gaussian Gram coercivity

Theorem 14.3 (Separated Gaussian Gram coercivity)**.** *Fix*

$$\alpha > 0, \qquad L > 0, \qquad \delta > 0.$$

Let

$$x_1, \ldots, x_N \in [-L, L]$$

be distinct points satisfying the separation condition

$$|x_j - x_k| \geq \delta \qquad (j \neq k).$$

Define

$$K_{jk} = e^{-\alpha(x_j - x_k)^2}.$$

Then there exists a constant

$$c(\alpha, L, \delta) > 0$$

such that

$$\lambda_{\min}(K) \geq c(\alpha, L, \delta)$$

for every such configuration.

Proof. Because the points lie in $[-L, L]$ and are δ-separated, the number of points is bounded:

$$N \leq N_{\max} := \left\lfloor \frac{2L}{\delta} \right\rfloor + 1.$$

For each fixed $N \leq N_{\max}$, consider the compact configuration space

$$\Omega_{N,L,\delta} = \{(x_1, \ldots, x_N) : -L \leq x_1 < \cdots < x_N \leq L, \quad x_{j+1} - x_j \geq \delta\}.$$

The map

$$(x_1, \ldots, x_N) \mapsto \lambda_{\min}(K(x_1, \ldots, x_N))$$

is continuous. For every configuration in $\Omega_{N,L,\delta}$, the Gaussian Gram matrix is strictly positive definite by the standard Gaussian kernel positivity argument. Hence

$$\lambda_{\min}(K) > 0$$

on $\Omega_{N,L,\delta}$.

Since $\Omega_{N,L,\delta}$ is compact, the continuous positive function

$$\lambda_{\min}(K)$$

attains a positive minimum:

$$c_N(\alpha, L, \delta) > 0.$$

Define

$$c(\alpha, L, \delta) := \min_{1 \leq N \leq N_{\max}} c_N(\alpha, L, \delta).$$

This minimum is positive because it is taken over finitely many positive constants. Therefore

$$\lambda_{\min}(K) \geq c(\alpha, L, \delta)$$

for all separated configurations. □

Remark 14.4. *This theorem gives the correct ordinary Gaussian Gram bound under a fixed separation hypothesis. It does not solve the zeta-side problem by itself, because local zero gaps are not expected to admit a height-uniform positive lower separation.*

14.3 Jet repair for near-collisions

When points nearly collide, the ordinary atoms

$$e^{ix_j t}$$

become nearly dependent. The correct limiting readout is not to keep several nearly identical atoms, but to pass to derivatives in the frequency parameter.

Indeed,

$$\frac{e^{i(x+\varepsilon)t} - e^{ixt}}{\varepsilon} \to it\, e^{ixt} \qquad (\varepsilon \to 0).$$

Thus a near-collision naturally produces a jet:

$$e^{iyt}, \quad (it)e^{iyt}, \quad (it)^2 e^{iyt}, \dots$$

Lemma 14.5 (Linear independence of exponential-polynomial jets)**.** *Let*

$$y_1, \dots, y_M \in \mathbb{R}$$

be distinct and let $m_1, \dots, m_M \in \mathbb{N}$*. If*

$$\sum_{r=1}^{M} \sum_{a=0}^{m_r-1} c_{r,a} (it)^a e^{iy_r t} = 0$$

for all real t*, then all coefficients* $c_{r,a}$ *vanish.*

Proof. The left-hand side is an exponential polynomial. Fix r. Apply the differential operator

$$\mathcal{D}_r = \prod_{s \neq r} \left(\frac{d}{dt} - iy_s \right)^{m_s}$$

to the identity. This annihilates every block with frequency y_s, $s \neq r$, and leaves an expression of the form

$$e^{iy_r t} P_r(t) = 0,$$

where P_r is a polynomial whose coefficients are linear combinations of

$$c_{r,0}, \dots, c_{r,m_r-1}.$$

Since $e^{iy_r t} \neq 0$, we get

$$P_r(t) = 0$$

for all real t. Hence all coefficients in this polynomial vanish. Repeating the argument for each r, and descending in degree if necessary, gives

$$c_{r,a} = 0$$

for all r, a. □

Theorem 14.6 (Confluent Gaussian Gram positivity)**.** *Let*

$$y_1, \dots, y_M \in \mathbb{R}$$

be distinct centers and let

$$m_1, \dots, m_M \in \mathbb{N}$$

be positive multiplicities. Define the jet functions

$$\phi_{r,a}(t) = (it)^a e^{iy_r t}, \qquad r = 1, \dots, M, \quad a = 0, \dots, m_r - 1.$$

Let the Gaussian weight be

$$w_\alpha(t) = e^{-t^2/(4\alpha)}.$$

Define the confluent Gaussian Gram matrix

$$G_{(r,a),(s,b)} = \int_{\mathbb{R}} \phi_{r,a}(t)\overline{\phi_{s,b}(t)} w_\alpha(t)\ \mathrm{d}t.$$

Then G is strictly positive definite.

Proof. For coefficients $c_{r,a}$, the associated quadratic form is

$$\sum_{r,a}\sum_{s,b} c_{r,a}\overline{c_{s,b}} G_{(r,a),(s,b)} = \int_{\mathbb{R}} \left| \sum_{r=1}^{M} \sum_{a=0}^{m_r-1} c_{r,a}(it)^a e^{iy_r t} \right|^2 w_\alpha(t)\ \mathrm{d}t.$$

Since

$$w_\alpha(t) > 0$$

for all real t, this integral can vanish only if

$$\sum_{r=1}^{M} \sum_{a=0}^{m_r-1} c_{r,a}(it)^a e^{iy_r t} = 0$$

for all real t. By the linear independence lemma for exponential-polynomial jets, all coefficients $c_{r,a}$ vanish. Hence the quadratic form is strictly positive on every nonzero coefficient vector, and G is strictly positive definite. □

Remark 14.7 (Interpretation)**.** *A near-collision of ordinary Gaussian atoms is not a reason to abandon coercivity. It means the readout basis must be refined. The correct limiting basis is a confluent jet basis. Ordinary Gram coercivity fails at collision scale; jet Gram positivity survives.*

14.4 Hybrid separation-or-jet target

The corrected target for the RH bridge is therefore not ordinary uniform Gaussian Gram coercivity for all clusters. The corrected target is a hybrid ordinary/confluent coercivity statement.

Open Problem 14.8 (Separation-or-jet Gaussian Gram coercivity)**.** *Let local unfolded zeta-zero clusters be decomposed into microclusters at a fixed scale $\delta > 0$.*

For centers separated by at least δ, use the ordinary Gaussian atoms

$$e^{iyt}.$$

For each microcluster of m near-colliding zeros, replace the nearly dependent atoms by the confluent jet system

$$e^{iyt}, \quad (it)e^{iyt}, \quad \ldots, \quad (it)^{m-1}e^{iyt}.$$

Prove a height-uniform lower bound for the resulting hybrid ordinary/confluent Gaussian Gram matrix:

$$\lambda_{\min}(G_{\text{hybrid}}) \geq c(\alpha, L, \delta, m_{\max}) > 0,$$

where $m_{\max}$ is the maximum allowed microcluster multiplicity inside the local window.

Remark 14.9 (Replacement for the old Gram bottleneck)**.** *The earlier open problem*

$$\lambda_{\min}(K) \geq c(\alpha, L) > 0$$

for ordinary Gaussian Gram matrices should be read as too strong unless a separation condition is added. The refined bottleneck is the separation-or-jet problem above.

Thus the Gram task in the complete proof list should be replaced by:

Prove height-uniform separation-or-jet Gaussian Gram coercivity for unfolded local zeta-zero clusters.

Remark 14.10 (Proof status)**.** *This section proves three facts:*

(i) ordinary uniform Gram coercivity fails without separation;

(ii) ordinary Gaussian Gram coercivity holds uniformly on compact separated configuration classes;

(iii) confluent jet Gaussian Gram matrices are strictly positive definite.

It does not yet prove the final height-uniform hybrid bound for genuine zeta-zero clusters. That hybrid bound remains the correct zeta-side analytic task.

15 Master Conditional Contradiction

Theorem 15.1 (Fixed-channel contradiction)**.** *Fix a candidate zero*

$$\rho = \beta + i\gamma, \qquad \beta \neq \frac{1}{2}.$$

Assume A1–A4 hold for the same localized channel associated with ρ. Then a contradiction follows. Therefore no off-critical zero can satisfy these four statements simultaneously.

Proof. Since

$$\beta \neq \frac{1}{2},$$

we have

$$a_\rho = \frac{\beta - \frac{1}{2}}{2\pi} \neq 0.$$

By A1 and the quartic lower bound, for sufficiently large τ,

$$P_4(\rho; \tau, L) \geq c_\rho \tau^4 L.$$

By A3 and A4,

$$Q^{(\rho)}_{\text{bank}}(f; \tau, L) \geq (\kappa - \eta(\tau)) P_4(\rho; \tau, L) - (C_0 + C_1)L.$$

For large τ, $\eta(\tau) \leq \kappa/2$. Hence

$$Q^{(\rho)}_{\text{bank}}(f; \tau, L) \geq \frac{\kappa}{2} P_4(\rho; \tau, L) - (C_0 + C_1)L.$$

Using the quartic lower bound,

$$Q^{(\rho)}_{\text{bank}}(f; \tau, L) \geq c' \tau^4 L - C'' L.$$

After subtracting the baseline $E^{(\rho)}_0 L$ and dividing by L, the right-hand side is positive for sufficiently large τ. Therefore

$$E^{(\rho)}_{\text{ex}}(f) > 0.$$

But A2, by localized closure, gives

$$E^{(\rho)}_{\text{ex}}(f) = 0.$$

This is a contradiction. □

Corollary 15.2 (Conditional RH)**.** *If A1–A4 hold for every off-critical candidate zero, then every nontrivial zero of $\zeta(s)$ lies on the critical line.*

Proof. If an off-critical zero existed, the fixed-channel contradiction theorem would apply to its localized channel. This is impossible. Therefore no off-critical zero exists. □

16 What a Complete Proof Would Require

A complete unconditional proof would require exactly the following seven tasks on the genuine zeta residue channel.

1. **Transported drift law.** Derive

$$\delta_\rho(u) = a_\rho u + O(1), \qquad a_\rho = \frac{\beta - \frac{1}{2}}{2\pi},$$

   from a concrete transport normalization of the explicit formula.

2. **Genuine localized witness coercivity.** Upgrade the windowed-packet coercivity theorem to the actual transported zero-channel.

3. **Localized no-hiding leakage.** Prove

$$Q^{(\rho)}_{\text{cross}} + Q^{(\rho)}_{\text{rest}} \geq -\eta(\tau)P_4 - C_1 L, \qquad \eta(\tau) \to 0.$$

4. **Bank-subcritical remainder.** Establish residue-side regularity strong enough to imply

$$Q^{(\rho)}_{\text{rest}} = o(P_4).$$

5. **Separation-or-jet Gaussian Gram coercivity.** Replace the naive ordinary Gram target by the corrected hybrid target: prove a height-uniform lower bound for separated ordinary modes together with confluent jet modes resolving near-collisions in local unfolded zeta-zero clusters.

6. **Localized exact window law.** Verify the localized exact window law for the concrete dipole-regularized channel arising from the explicit formula.

7. **Model-to-zeta transfer.** Transfer the model estimates in the appendices to the genuine zeta residue channel.

These seven tasks are explicit. They are not claimed to be solved here.

17 Reality Check

17.1 Proved inside this document

The following statements are proved here:

- localized exact window law implies $E^{(\rho)}_{\text{ex}}(f) = 0$;
- transported drift implies the quartic lower bound $P_4(\rho; \tau, L) \gtrsim \tau^4 L$;
- localized witness coercivity plus no-hiding leakage implies the bridge lower bound;
- a C^2 remainder with exponent $m_0 < 2$ is bank-subcritical;
- finite Gaussian Gram matrices of distinct ordinates are positive definite;
- pure exponential modes satisfy localized coercivity in a controlled model;
- windowed pure-exponential packets satisfy localized coercivity in a controlled model;
- coercivity is stable under small bank-energy perturbations;
- cross terms are quartic-subcritical under nonstationary phase assumptions;

- Gaussian heat-semigroup regularization sends tempered singular readouts to smooth profiles;
- the Source–Love Channel is a strict contraction meta-model with a unique fixed point.
- ordinary uniform Gaussian Gram coercivity without separation is impossible;
- separated Gaussian Gram coercivity holds by compactness;
- confluent Gaussian jet Gram matrices are strictly positive definite;

17.2 Still open

The seven zeta-side tasks listed in the previous section remain open.

Remark 17.1. *The document is designed to be falsifiable at the bottlenecks. Failure of drift, coercivity, no-hiding, Gram control, or exact window closure on the genuine channel would block the RH implication.*

18 Outlook and Companion Status

The symbolic Source–Love geometry is useful for exposition:

$$\text{Source} = \text{drift-generating deviation}, \qquad \text{Love} = \text{closure-enforcing contraction},$$

and

$$\text{critical line} = \text{compatibility interface}.$$

This symbolic layer is not part of the proof chain. The detailed symbolic geometry belongs in a companion note, not in the main RH checkpoint.

The formal proof chain in this document remains exactly:

$$\text{transported drift}+\text{witness coercivity}+\text{no-hiding leakage}+\text{exact window} \Longrightarrow \text{fixed-channel contradiction}.$$

19 Conclusion

The localized transport–spectral program reduces the Riemann Hypothesis to a clean conditional contradiction.

The closure side forces

$$E_{\mathrm{ex}}^{(\rho)}(f) = 0.$$

The obstruction side, under transported drift and no-hiding, forces

$$E_{\mathrm{ex}}^{(\rho)}(f) > 0.$$

The contradiction is valid once the localized bridge package is verified on the genuine zeta residue channel.

This v3.2 checkpoint isolates the problem into explicit analytic bottlenecks and supplies rigorous model bricks showing that the required mechanisms are mathematically nonempty. The remaining work is hard, but it is now localized, modular, and precise.

Closure kills leakage.
Obstruction creates drift.
No-Hiding is the lock.
Localization makes the key visible.

Architecture is not proof.
But a sharp architecture tells us exactly what the proof must still accomplish.

A Soft Survival of a Transported Readout

This appendix records a structural compatibility lemma. Soft regularization does not create a transported drift, but it can prevent a pre-existing transported signal from being annihilated by a local hard readout collapse.

Lemma A.1 (Soft phase survival above baseline)**.** *Let $U \subset \mathbb{R}$ be a neighborhood of a local singular point. Let $\varepsilon > 0$ and define*

$$r_\varepsilon(\xi) = \sqrt{\xi^2 + \varepsilon^2}.$$

Assume that the effective readout admits an expansion

$$\Pi_{\eta,\varepsilon}(\xi) = \Pi_0 + \eta W(r_\varepsilon(\xi)) + R_{\eta,\varepsilon}(\xi), \qquad \xi \in U,$$

where:

(i) $\Pi_0 \in \mathbb{R}$*;*

(ii) $W : [0,\infty) \to [0,\infty)$ *is continuous and nonnegative;*

(iii) there exists $C > 0$ such that

$$|R_{\eta,\varepsilon}(\xi)| \le C\eta^2$$

for all $\xi \in U$ and sufficiently small $\eta > 0$;

(iv) there exists $w_* > 0$ such that

$$W(r_\varepsilon(\xi)) \ge w_*$$

for all $\xi \in U$.

Then there exists $\eta_0 > 0$ such that for all $0 < \eta < \eta_0$,

$$\Pi_{\eta,\varepsilon}(\xi) \ge \Pi_0 + \frac{1}{2}\eta w_* > \Pi_0$$

for all $\xi \in U$. If $\Pi_0 = 0$, then

$$\Pi_{\eta,\varepsilon}(\xi) > 0.$$

Proof. Using the expansion,

$$\Pi_{\eta,\varepsilon}(\xi) = \Pi_0 + \eta W(r_\varepsilon(\xi)) + R_{\eta,\varepsilon}(\xi).$$

By the lower bound on W and the remainder estimate,

$$\Pi_{\eta,\varepsilon}(\xi) \ge \Pi_0 + \eta w_* - C\eta^2.$$

Choose $\eta_0 > 0$ such that

$$C\eta^2 \le \frac{1}{2}\eta w_*$$

for $0 < \eta < \eta_0$. Then

$$\Pi_{\eta,\varepsilon}(\xi) \ge \Pi_0 + \frac{1}{2}\eta w_*.$$

If $\Pi_0 = 0$, this is strictly positive. □

Corollary A.2 (Compatibility with transported drift)**.** *Let f_ρ be a localized transported mode associated with an off-critical zero*

$$\rho = \beta + i\gamma, \qquad \beta \ne \frac{1}{2}.$$

Suppose that in a transported frame its effective phase variable satisfies

$$\delta_\rho(u) = a_\rho u + r_\rho(u), \qquad a_\rho = \frac{\beta - \frac{1}{2}}{2\pi},$$

with r_ρ bounded. If the local readout near a hard singularity satisfies the soft expansion in the preceding lemma, then local loss of hard sign-readout does not force annihilation of the transported signal in the soft channel.

Proof. Apply the previous lemma to the local readout of the transported channel. □

Remark A.3. *This result does not prove the transported drift law for the genuine zeta function. It proves only that if a transported drift is already present, soft regularization can keep it visible through a local hard readout collapse.*

B Pure Exponential Coercivity Model

This appendix proves localized witness coercivity in the exact pure-exponential model.

Proposition B.1 (Localized witness coercivity: pure exponential model)**.** *Let*

$$\rho = \beta + i\gamma, \qquad 0 < \beta < 1, \qquad \beta \neq \frac{1}{2}.$$

Define

$$g_\rho(u) = A_\rho e^{\rho u}, \qquad A_\rho = \exp\left(\frac{\rho^2}{4\pi}\right).$$

Assume the transported drift law. Let $w_{\rho,\tau,L}$ be a plateau window satisfying

$$0 < m \leq w_{\rho,\tau,L}(u) \leq M < \infty$$

for all $u \in [\tau, \tau + L]$. Then there exist constants

$$\kappa > 0, \qquad C_0 \geq 0, \qquad \tau_0 > 0$$

such that for all $\tau \geq \tau_0$ and $L \geq 1$,

$$Q^{(\rho)}_{\text{bank}}(g_\rho; \tau, L) \geq \kappa P_4(\rho; \tau, L) - C_0 L.$$

Proof. Since g_ρ is a pure exponential,

$$W_{a_j} g_\rho(u) = g_\rho(u + a_j) - 2g_\rho(u) + g_\rho(u - a_j) = d_{\rho,a_j} g_\rho(u),$$

where

$$d_{\rho,a_j} = e^{\rho a_j} - 2 + e^{-\rho a_j}.$$

Hence

$$Q^{(\rho)}_{\text{bank}}(g_\rho; \tau, L) = |A_\rho|^2 \sum_{j=0}^{J} w_j^2 |d_{\rho,a_j}|^2 \int_\tau^{\tau+L} w_{\rho,\tau,L}(u) e^{2\beta u}\, \mathrm{d}u.$$

Set

$$S_\rho = \sum_{j=0}^{J} w_j^2 |d_{\rho,a_j}|^2.$$

We claim $S_\rho > 0$. If $d_{\rho,a_j} = 0$, then

$$e^{\rho a_j} + e^{-\rho a_j} = 2,$$

so

$$\cosh(\rho a_j) = 1.$$

The solutions of $\cosh z = 1$ are $z = 2\pi i k$, $k \in \mathbb{Z}$, hence have zero real part. But

$$\Re(\rho a_j) = \beta a_j > 0.$$

Thus $d_{\rho,a_j} \neq 0$, and therefore $S_\rho > 0$.

Using $w_{\rho,\tau,L} \geq m$,

$$Q^{(\rho)}_{\text{bank}}(g_\rho;\tau,L) \geq |A_\rho|^2 S_\rho m \int_\tau^{\tau+L} e^{2\beta u}\,\mathrm{d}u.$$

Therefore

$$Q^{(\rho)}_{\text{bank}}(g_\rho;\tau,L) \geq c_1\left(e^{2\beta(\tau+L)} - e^{2\beta\tau}\right),$$

where

$$c_1 = |A_\rho|^2 S_\rho \frac{m}{2\beta} > 0.$$

By the polynomial upper bound for P_4,

$$P_4(\rho;\tau,L) \leq C_\rho\left((\tau+L)^5 - \tau^5 + L\right).$$

The exponential lower bound dominates the polynomial upper bound for sufficiently large τ, uniformly for $L \geq 1$. Hence, choosing $\kappa > 0$ small and τ_0 large,

$$Q^{(\rho)}_{\text{bank}}(g_\rho;\tau,L) \geq \kappa P_4(\rho;\tau,L) - C_0 L.$$

□

Remark B.2. *This model is intentionally idealized. It does not prove coercivity for the genuine zeta residue channel.*

C Windowed Packet Coercivity and Perturbative Stability

This appendix is the main technical upgrade beyond the pure exponential model. It shows that coercivity survives localization by a packet window with an interior plateau.

Let

$$A_* = \max_{0\leq j\leq J} a_j.$$

Definition C.1 (Admissible packet window)**.** *An admissible packet window $\chi_{\tau,L} \in C^2(\mathbb{R})$ satisfies:*

$$0 \leq \chi_{\tau,L} \leq 1,$$

$$\chi_{\tau,L}(u) = 1 \quad \textit{for } u \in [\tau + A_*,\, \tau + L - A_*],$$

and

$$\operatorname{supp}\chi_{\tau,L} \subset [\tau, \tau+L].$$

We assume $L \geq 2A_* + 1$, so the interior plateau has length at least 1.

Definition C.2 (Windowed pure-exponential packet)**.** *Define*

$$h_{\rho,\tau,L}(u) = \chi_{\tau,L}(u) A_\rho e^{\rho u}.$$

Lemma C.3 (Interior exactness of the second-difference witness)**.** *If*

$$u \in [\tau + A_*,\, \tau + L - A_*],$$

then for every bank scale a_j,

$$W_{a_j} h_{\rho,\tau,L}(u) = d_{\rho,a_j} A_\rho e^{\rho u},$$

where

$$d_{\rho,a_j} = e^{\rho a_j} - 2 + e^{-\rho a_j}.$$

Proof. If

$$u \in [\tau + A_*,\, \tau + L - A_*],$$

then

$$u - a_j,\ u,\ u + a_j \in [\tau, \tau + L]$$

and, more precisely, all three points lie inside the plateau region where

$$\chi_{\tau,L} = 1.$$

Therefore

$$h_{\rho,\tau,L}(u \pm a_j) = A_\rho e^{\rho(u \pm a_j)}$$

and

$$h_{\rho,\tau,L}(u) = A_\rho e^{\rho u}.$$

Hence

$$W_{a_j} h_{\rho,\tau,L}(u) = A_\rho e^{\rho u}(e^{\rho a_j} - 2 + e^{-\rho a_j}) = d_{\rho,a_j} A_\rho e^{\rho u}.$$

□

Theorem C.4 (Localized packet coercivity)**.** *Let*

$$\rho = \beta + i\gamma, \qquad 0 < \beta < 1, \qquad \beta \neq \frac{1}{2}.$$

Assume the transported drift law. Let $h_{\rho,\tau,L}$ be the windowed packet above, with $L \geq 2A_* + 1$. Then there exist constants

$$\kappa > 0, \qquad C_0 \geq 0, \qquad \tau_0 > 0$$

such that for all $\tau \geq \tau_0$,

$$Q^{(\rho)}_{\text{bank}}(h_{\rho,\tau,L}; \tau, L) \geq \kappa P_4(\rho; \tau, L) - C_0 L.$$

Proof. Because the integrand in the bank energy is nonnegative, we may restrict the integral to the interior plateau:

$$Q^{(\rho)}_{\text{bank}}(h_{\rho,\tau,L}; \tau, L) \geq \sum_{j=0}^{J} w_j^2 \int_{\tau+A_*}^{\tau+L-A_*} w_\rho(u) |W_{a_j} h_{\rho,\tau,L}(u)|^2 \, \mathrm{d}u.$$

Assume the localization weight is bounded below on the relevant packet region:

$$w_\rho(u) \geq m_\rho > 0$$

on the interior plateau. Using interior exactness,

$$|W_{a_j} h_{\rho,\tau,L}(u)|^2 = |d_{\rho,a_j}|^2 |A_\rho|^2 e^{2\beta u}.$$

Therefore

$$Q^{(\rho)}_{\text{bank}}(h_{\rho,\tau,L}; \tau, L) \geq m_\rho |A_\rho|^2 \left(\sum_{j=0}^{J} w_j^2 |d_{\rho,a_j}|^2 \right) \int_{\tau+A_*}^{\tau+L-A_*} e^{2\beta u} \, \mathrm{d}u.$$

As in the pure exponential model,

$$S_\rho = \sum_{j=0}^{J} w_j^2 |d_{\rho,a_j}|^2 > 0.$$

Hence

$$Q^{(\rho)}_{\text{bank}}(h_{\rho,\tau,L}; \tau, L) \geq C_\rho \int_{\tau+A_*}^{\tau+L-A_*} e^{2\beta u} \, \mathrm{d}u$$

for some $C_\rho > 0$.

The plateau has length at least 1, so the exponential integral is positive. More explicitly,

$$\int_{\tau+A_*}^{\tau+L-A_*} e^{2\beta u}\,\mathrm{d}u = \frac{e^{2\beta(\tau+L-A_*)} - e^{2\beta(\tau+A_*)}}{2\beta}.$$

This exponential quantity dominates the polynomial upper bound

$$P_4(\rho;\tau,L) \le C\left((\tau+L)^5 - \tau^5 + L\right)$$

for sufficiently large τ, uniformly for $L \ge 2A_* + 1$. Therefore, choosing $\kappa > 0$ small and τ_0 large enough,

$$Q^{(\rho)}_{\mathrm{bank}}(h_{\rho,\tau,L};\tau,L) \ge \kappa P_4(\rho;\tau,L) - C_0 L.$$

□

Remark C.5. *The proof does not need to estimate boundary layers from below. Since the bank energy is nonnegative, the boundary contribution can be ignored. The coercive contribution comes entirely from the interior plateau where the second-difference witness is exact.*

Lemma C.6 (Quadratic lower stability)**.** *For all $x, y \in \mathbb{C}$,*

$$|x+y|^2 \ge \frac{1}{2}|x|^2 - |y|^2.$$

Proof. Using

$$|x| = |(x+y) - y| \le |x+y| + |y|,$$

we have

$$|x|^2 \le 2|x+y|^2 + 2|y|^2.$$

Rearranging gives

$$|x+y|^2 \ge \frac{1}{2}|x|^2 - |y|^2.$$

□

Proposition C.7 (Perturbative stability of coercivity)**.** *Suppose a model mode g satisfies*

$$Q^{(\rho)}_{\mathrm{bank}}(g;\tau,L) \ge \kappa P_4(\rho;\tau,L) - C_0 L.$$

Let e be a perturbation satisfying

$$Q^{(\rho)}_{\mathrm{bank}}(e;\tau,L) \le \frac{\kappa}{4} P_4(\rho;\tau,L) + C_e L.$$

Then

$$Q^{(\rho)}_{\mathrm{bank}}(g+e;\tau,L) \ge \frac{\kappa}{4} P_4(\rho;\tau,L) - \left(\frac{C_0}{2} + C_e\right) L.$$

Proof. Apply the quadratic lower stability inequality pointwise with

$$x = W_{a_j} g(u), \qquad y = W_{a_j} e(u).$$

After multiplying by $w_j^2 w_\rho(u)$, summing over j, and integrating,

$$Q^{(\rho)}_{\mathrm{bank}}(g+e;\tau,L) \ge \frac{1}{2} Q^{(\rho)}_{\mathrm{bank}}(g;\tau,L) - Q^{(\rho)}_{\mathrm{bank}}(e;\tau,L).$$

Using the two hypotheses,

$$Q^{(\rho)}_{\mathrm{bank}}(g+e;\tau,L) \ge \frac{1}{2}(\kappa P_4 - C_0 L) - \left(\frac{\kappa}{4} P_4 + C_e L\right).$$

Thus

$$Q^{(\rho)}_{\mathrm{bank}}(g+e;\tau,L) \ge \frac{\kappa}{4} P_4 - \left(\frac{C_0}{2} + C_e\right) L.$$

□

Remark C.8. *This identifies the next genuine zeta-side target: prove that the difference between the actual transported zero-channel and a controlled packet model is small in bank energy relative to P_4.*

D Nonstationary Cross-Term Decay

This appendix proves a model nonstationary-phase estimate for cross-term decay.

Lemma D.1 (Localized cross-term decay by nonstationary phase)**.** *Define*

$$Q^{(\rho)}_{\mathrm{cross}}(\tau, L) = 2\sum_{j=0}^{J} w_j^2 \operatorname{Re}\int_{\tau}^{\tau+L} w_\rho(u) B_{\rho,j}(u)\overline{B_{\mathrm{rest},j}(u)} e^{i\Phi_j(u)}\,\mathrm{d}u.$$

Assume that, on $[\tau, \tau+L]$,

$$\|w_\rho\|_\infty + \|w_\rho'\|_\infty \le C_w,$$
$$|B_{\rho,j}(u)| + |B'_{\rho,j}(u)| \le C_\rho \tau^2,$$
$$|B_{\mathrm{rest},j}(u)| + |B'_{\mathrm{rest},j}(u)| \le C_{\mathrm{rest}},$$
$$|\Phi_j'(u)| \ge c_\Phi \tau, \qquad |\Phi_j''(u)| \le C_\Phi.$$

Then

$$|Q^{(\rho)}_{\mathrm{cross}}(\tau, L)| \le C_{\mathrm{cross}} \tau L$$

for sufficiently large τ *and all* $L \ge 1$.

Proof. Set

$$g_j(u) = w_\rho(u) B_{\rho,j}(u)\overline{B_{\mathrm{rest},j}(u)}.$$

By the product rule and the assumed bounds,

$$|g_j(u)| + |g_j'(u)| \le C_g \tau^2.$$

Let

$$I_j = \int_{\tau}^{\tau+L} g_j(u) e^{i\Phi_j(u)}\,\mathrm{d}u.$$

Since Φ_j' is bounded away from zero,

$$e^{i\Phi_j(u)} = \frac{1}{i\Phi_j'(u)} \frac{\mathrm{d}}{\mathrm{d}u} e^{i\Phi_j(u)}.$$

Integration by parts gives

$$I_j = \left[\frac{g_j(u)}{i\Phi_j'(u)} e^{i\Phi_j(u)}\right]_{\tau}^{\tau+L} - \int_{\tau}^{\tau+L}\left(\frac{g_j'(u)}{i\Phi_j'(u)} - \frac{g_j(u)\Phi_j''(u)}{i(\Phi_j'(u))^2}\right) e^{i\Phi_j(u)}\,\mathrm{d}u.$$

Taking absolute values,

$$|I_j| \le 2\sup \frac{|g_j|}{|\Phi_j'|} + \int_{\tau}^{\tau+L} \frac{|g_j'|}{|\Phi_j'|}\,\mathrm{d}u + \int_{\tau}^{\tau+L} \frac{|g_j||\Phi_j''|}{|\Phi_j'|^2}\,\mathrm{d}u.$$

Using the hypotheses,

$$\frac{|g_j|}{|\Phi_j'|} \le C\tau,$$
$$\frac{|g_j'|}{|\Phi_j'|} \le C\tau,$$

and

$$\frac{|g_j||\Phi_j''|}{|\Phi_j'|^2} \le C.$$

Thus

$$|I_j| \le C\tau + C\tau L + CL \le C'\tau L$$

for $\tau \ge 1$, $L \ge 1$. Summing over finitely many j gives the result. □

Corollary D.2 (Cross term is quartic-subcritical)**.** *If*

$$P_4(\rho;\tau,L) \geq c_4\tau^4 L,$$

then

$$\frac{|Q^{(\rho)}_{\text{cross}}(\tau,L)|}{P_4(\rho;\tau,L)} \leq \frac{C_{\text{cross}}}{c_4\tau^3} \to 0.$$

Thus

$$Q^{(\rho)}_{\text{cross}} = o(P_4).$$

Proof. Combine

$$|Q^{(\rho)}_{\text{cross}}(\tau,L)| \leq C_{\text{cross}}\tau L$$

with

$$P_4(\rho;\tau,L) \geq c_4\tau^4 L.$$

□

E Gaussian Hammer: Heat-Semigroup Regularization

This appendix records the Gaussian Hammer principle: Gaussian readout converts singular or distributional data into a smooth effective profile and suppresses high frequencies exponentially.

We use the Fourier convention

$$\widehat{f}(\xi) = \int_{\mathbb{R}^n} f(x)e^{-2\pi i x\cdot\xi}\,\mathrm{d}x.$$

For $\sigma > 0$, define

$$G_\sigma(x) = \sigma^{-n}e^{-\pi|x|^2/\sigma^2}.$$

Lemma E.1 (Fourier transform of the scaled Gaussian)**.** *With the above convention,*

$$\widehat{G_\sigma}(\xi) = e^{-\pi\sigma^2|\xi|^2}.$$

Proof. The standard identity is

$$\widehat{e^{-\pi|x|^2}}(\xi) = e^{-\pi|\xi|^2}.$$

Since

$$G_\sigma(x) = \sigma^{-n}G_1(x/\sigma),$$

the Fourier scaling rule gives

$$\widehat{G_\sigma}(\xi) = \widehat{G_1}(\sigma\xi) = e^{-\pi\sigma^2|\xi|^2}.$$

□

Theorem E.2 (Gaussian hammer regularizes singular readout)**.** *Let $u \in \mathcal{S}'(\mathbb{R}^n)$ be a tempered distribution and define*

$$u_\sigma = G_\sigma * u.$$

Then:

(1) $u_\sigma \in C^\infty(\mathbb{R}^n)$*;*

(2)

$$\widehat{u_\sigma}(\xi) = e^{-\pi\sigma^2|\xi|^2}\widehat{u}(\xi);$$

(3) high-frequency components are exponentially suppressed by

$$e^{-\pi\sigma^2|\xi|^2};$$

(4) if $u = \delta_{x_0}$, then

$$u_\sigma(x) = G_\sigma(x - x_0).$$

Proof. Since $G_\sigma \in \mathcal{S}(\mathbb{R}^n)$ and $u \in \mathcal{S}'(\mathbb{R}^n)$, convolution with G_σ produces a smooth function. For every multiindex α,

$$\partial^\alpha u_\sigma = \partial^\alpha (G_\sigma * u) = (\partial^\alpha G_\sigma) * u,$$

which is well-defined and continuous.

The Fourier convolution identity gives

$$\widehat{u_\sigma} = \widehat{G_\sigma}\widehat{u} = e^{-\pi\sigma^2|\xi|^2}\widehat{u}.$$

The exponential multiplier suppresses high frequencies.

Finally,

$$G_\sigma * \delta_{x_0} = G_\sigma(\cdot - x_0),$$

so

$$u_\sigma(x) = G_\sigma(x - x_0).$$

□

Corollary E.3 (Gaussian suppression beats every polynomial)**.** *For every $N > 0$, there exists $C_{N,\sigma} > 0$ such that*

$$e^{-\pi\sigma^2|\xi|^2} \le C_{N,\sigma}(1 + |\xi|)^{-N}.$$

Proof. Let $r = |\xi|$. The function

$$r \mapsto e^{-\pi\sigma^2 r^2}(1 + r)^N$$

is continuous on $[0, \infty)$ and tends to 0 as $r \to \infty$. Hence it is bounded. □

Proposition E.4 (Gaussian readout as heat flow)**.** *Let Δ be the Laplacian on $\mathbb{R}^n$. Then*

$$G_\sigma * u = e^{t\Delta}u, \qquad t = \frac{\sigma^2}{4\pi}.$$

Proof. The Fourier transform of the Laplacian is

$$\widehat{\Delta u}(\xi) = -4\pi^2|\xi|^2\widehat{u}(\xi).$$

Thus

$$\widehat{e^{t\Delta}u}(\xi) = e^{-4\pi^2 t|\xi|^2}\widehat{u}(\xi).$$

Choosing

$$t = \frac{\sigma^2}{4\pi}$$

gives

$$e^{-4\pi^2 t|\xi|^2} = e^{-\pi\sigma^2|\xi|^2},$$

which is exactly the Fourier multiplier for convolution by G_σ. □

Remark E.5. *This does not prove removal of physical singularities in general relativity. It proves the analytic readout statement:*

$$\textit{singular raw source} \longmapsto \textit{smooth Gaussian effective profile.}$$

F Source–Love Channel: Companion Meta-Model

This appendix is not used as an input to the RH contradiction. It records a separate operator model for the general principle that stable structure can arise from the interaction of transport, controlled perturbation, and contraction.

Let

$$\mathcal{H} \simeq \mathbb{C}^d,$$

and let $\mathcal{D}(\mathcal{H})$ denote the set of density operators on $\mathcal{H}$. Fix:

- a unitary operator $U : \mathcal{H} \to \mathcal{H}$;
- a target state $\rho_\infty \in \mathcal{D}(\mathcal{H})$;
- a source state $\rho_F \in \mathcal{D}(\mathcal{H})$;
- parameters
$$\lambda \in (0, 1], \qquad \gamma \in [0, 1).$$

Define the Source–Love Channel by

$$\mathcal{T}(\rho) = (1-\lambda)(1-\gamma)U\rho U^\dagger + (1-\lambda)\gamma U\rho_F U^\dagger + \lambda\rho_\infty.$$

The three terms encode:

$$\text{transport} + \text{source perturbation} + \text{stabilizing contraction}.$$

Theorem F.1 (Source–Love contraction)**.** *For all $\rho, \sigma \in \mathcal{D}(\mathcal{H})$,*

$$\|\mathcal{T}(\rho) - \mathcal{T}(\sigma)\|_1 = (1-\lambda)(1-\gamma)\|\rho - \sigma\|_1.$$

Hence $\mathcal{T}$ is a strict contraction whenever

$$q = (1-\lambda)(1-\gamma) < 1.$$

Proof. Subtract:

$$\mathcal{T}(\rho) - \mathcal{T}(\sigma) = (1-\lambda)(1-\gamma)U(\rho - \sigma)U^\dagger.$$

The trace norm is unitarily invariant:

$$\|U(\rho - \sigma)U^\dagger\|_1 = \|\rho - \sigma\|_1.$$

Therefore

$$\|\mathcal{T}(\rho) - \mathcal{T}(\sigma)\|_1 = (1-\lambda)(1-\gamma)\|\rho - \sigma\|_1.$$

□

Corollary F.2 (Unique stable attractor)**.** *The map $\mathcal{T}$ has a unique fixed point $\rho_* \in \mathcal{D}(\mathcal{H})$, and*

$$\|\mathcal{T}^n(\rho) - \rho_*\|_1 \le q^n\|\rho - \rho_*\|_1.$$

Proof. This is the Banach fixed-point theorem applied to the complete metric space

$$(\mathcal{D}(\mathcal{H}), \|\cdot\|_1).$$

□

F.1 Fixed point via Neumann series

Let

$$c = (1-\lambda)(1-\gamma),$$

and define

$$b = (1-\lambda)\gamma U \rho_F U^\dagger + \lambda \rho_\infty.$$

Then the fixed point equation is

$$\rho_* = cU\rho_* U^\dagger + b.$$

Equivalently,

$$(I - c\,\mathrm{Ad}_U)\rho_* = b,$$

where

$$\mathrm{Ad}_U(X) = UXU^\dagger.$$

Proposition F.3 (Neumann-series representation)**.** *The unique fixed point satisfies*

$$\rho_* = \sum_{n=0}^{\infty} c^n\,\mathrm{Ad}_U^n(b).$$

Proof. Since

$$\|\,\mathrm{Ad}_U\,\|_{1\to 1} = 1$$

and $0 \le c < 1$, the operator $I - c\,\mathrm{Ad}_U$ is invertible by the Neumann series:

$$(I - c\,\mathrm{Ad}_U)^{-1} = \sum_{n=0}^{\infty} c^n\,\mathrm{Ad}_U^n\,.$$

Applying this to b gives the result. □

F.2 Progress as reduction of epistemic debt

Define the epistemic debt

$$D(\rho) = \|\rho - \rho_*\|_1.$$

Then

$$D(\mathcal{T}\rho) = qD(\rho).$$

Definition F.4 (Progress functional)**.** *Define*

$$\mathcal{P}(\rho) = D(\rho) - D(\mathcal{T}\rho).$$

Hence

$$\mathcal{P}(\rho) = (1-q)D(\rho).$$

Theorem F.5 (Progress is positive away from the attractor)**.** *If $\rho \neq \rho_*$, then*

$$\mathcal{P}(\rho) > 0.$$

If $\rho = \rho_*$, then

$$\mathcal{P}(\rho) = 0.$$

Proof. Since $1 - q > 0$,

$$\mathcal{P}(\rho) = (1-q)\|\rho - \rho_*\|_1.$$

This is positive exactly when $\rho \neq \rho_*$. □

F.3 Continuous-time limit

Let

$$\lambda = \kappa\Delta t, \qquad \gamma = \mu\Delta t, \qquad U = e^{-iH\Delta t}.$$

Then

$$U\rho U^\dagger = \rho - i\Delta t[H,\rho] + o(\Delta t).$$

Proposition F.6 (Continuous-time Lindblad-type limit)**.** *In the limit $\Delta t \to 0$, the Source–Love dynamics converge formally to*

$$\frac{\mathrm{d}\rho}{\mathrm{d}t} = -i[H,\rho] + \kappa(\rho_\infty - \rho) + \mu(\rho_F - \rho).$$

Proof. Substitute

$$\lambda = \kappa\Delta t, \qquad \gamma = \mu\Delta t, \qquad U = e^{-iH\Delta t}$$

into $\mathcal{T}$. Keeping only first-order terms gives

$$\mathcal{T}(\rho) - \rho = \Delta t\,(-i[H,\rho] + \kappa(\rho_\infty - \rho) + \mu(\rho_F - \rho)) + o(\Delta t).$$

Dividing by Δt and taking $\Delta t \to 0$ gives the generator. □

Remark F.7. *The Source–Love Channel is not part of the RH proof chain. It is a companion model for the general principle that stable readout can arise from coherent transport, controlled injection, and contraction.*

Progress is not doing everything possible.

Progress is converting epistemic debt into stable readout.

Love is not sentiment; it is the contraction that lets exploration survive without exploding.